Distinguishing Sepsis from Cytokine Release Syndrome After CAR-T Cell Therapy: A Machine Learning Approach Using Real-World EHR Data

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BACKGROUND

- Cytokine release syndrome (CRS) is an expected complication of chimeric antigen receptor T-cell (CAR-T) therapy. It presents with fever, hypotension, elevated inflammatory markers, and organ dysfunction, exactly as sepsis does.
- Management diverges sharply. CRS is treated with IL-6 blockade and supportive care; sepsis requires broad-spectrum antibiotics, source control, and resuscitation.
- Misdiagnosis is costly in both directions: treating CRS as sepsis wastes antimicrobials and delays immunosuppression, while treating sepsis as CRS delays life-saving therapy.
- Cultures take 24–72 hours to result, and no validated rapid decision-support tool exists for this differential.

OBJECTIVE

To develop and evaluate machine learning models that distinguish sepsis from CRS in post-CAR-T patients using routinely available EHR data recorded before the clinical diagnosis is made.

METHODS

- **Data source.** Retrospective observational analysis of the de-identified UC San Diego Health OMOP Common Data Model, linked to the OHDSI standardized vocabulary.
- **Cohort.** 1,821 CAR-T recipients identified from procedure records. Events, labs, and diagnoses collected from inpatient encounters within 30 days of infusion.
- **Labels.** Diagnosis codes classified patients as sepsis (n=80) or CRS-only (n=395); overlapping diagnoses were labeled sepsis. Labeled cohort: 475 patients, 16.8% sepsis.
- **Features.** 30 predictors: binary diagnostic indicators, post-infusion lab statistics (WBC, CRP, lactate, NLR), pre-infusion baseline labs and deltas, and comorbidities.
- **Temporal integrity.** Every feature was derived exclusively from data recorded before the diagnosis date, simulating deployment at the point of clinical uncertainty.
- **Models.** Logistic regression, random forest, gradient-boosted trees, and a neural network, each class-weighted and standardized, evaluated by 70/30 holdout and 5-fold stratified cross-validation. Random forest and gradient boosting combine hundreds of decision trees, each a branching sequence of yes/no splits on clinical variables, so that the errors of any single tree are averaged out.
- **Analytic sets.** Cross-validated results use all 475 labeled patients. The 70/30 holdout uses 465 (80 sepsis, 385 CRS) after 10 CRS patients without qualifying clinical events dropped out during feature construction.

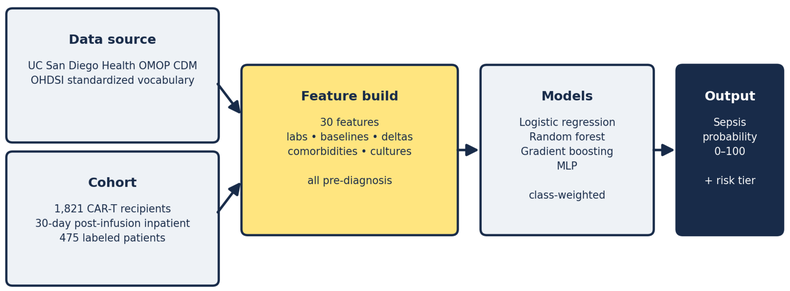


Figure 1. Data flow from the OMOP CDM through feature construction to a per-patient sepsis probability.

RESULTS

0.85	0.84	475
AUC-ROC, holdout	AUC-ROC, 5-fold CV	labeled patients

AUC-ROC: how often the model scores a sepsis patient above a CRS patient (0.50 = chance).
AUC-PR: how often its sepsis flags are correct across thresholds (0.17 = chance at this prevalence).

- **Random forest discriminated best.** AUC-ROC 0.85 on the 465-patient holdout and 0.84 cross-validated across all 475; AUC-PR 0.57 and 0.49 against a 0.17 prevalence baseline, roughly three times chance.
- **At a 0.50 threshold,** random forest reached 82.5% accuracy, 60.0% sensitivity (48/80 sepsis cases), 87.1% specificity (344/395 CRS cases), and 48.5% precision.
- **The models trade off differently.** Gradient-boosted trees reached the highest accuracy (83.6%) but detected only 33.8% of sepsis; logistic regression reached 83.8% sensitivity at 31.2% precision.
- **The neural network failed.** With 475 patients and 80 positive cases, the multilayer perceptron performed near chance (AUC-ROC 0.54).

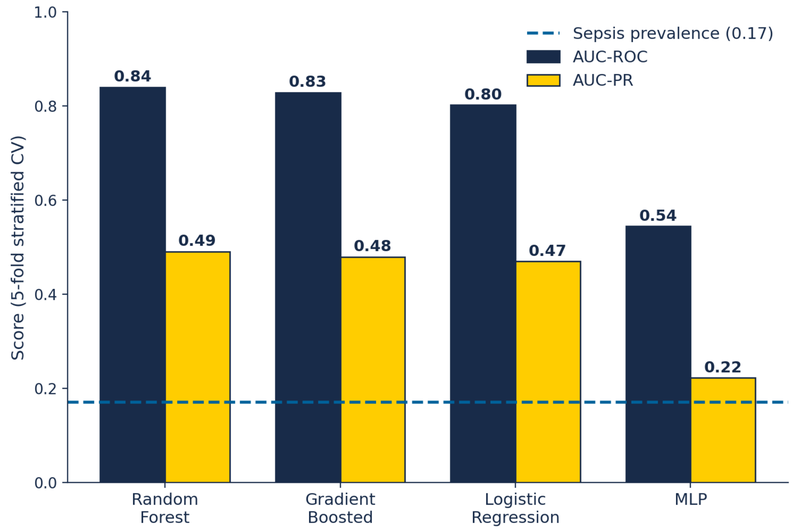


Figure 2. Discrimination by model, 5-fold stratified cross-validation. AUC-PR is the more honest metric under 16.8% prevalence.

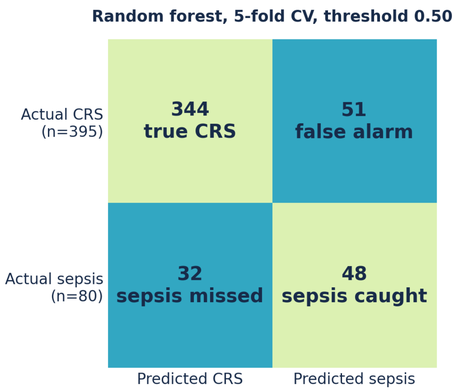


Figure 3. Random forest confusion matrix. The 32 missed sepsis cases are the clinically critical error.

PREDICTORS

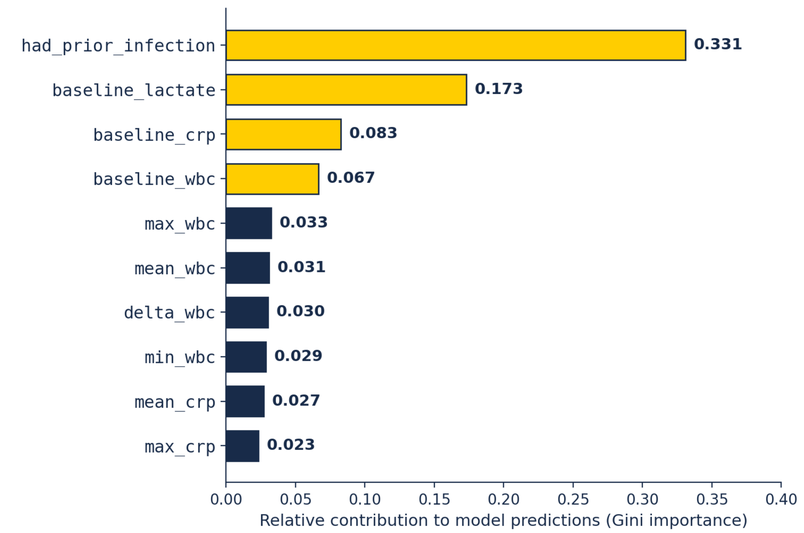


Figure 4. Top 10 random forest features by Gini importance. Pre-infusion history and baseline labs dominate.

- **Prior infection** in the year before infusion was the single strongest predictor (Gini 0.331), consistent with immunocompromise, colonization, and predisposition to infection under further immunosuppression.
- **Baseline lactate and CRP** (0.173, 0.083) suggest pre-existing tissue hypoperfusion or subclinical inflammation lowers the threshold for sepsis.
- **Rising WBC from baseline** favored sepsis, while high WBC variability favored CRS, matching the waxing-and-waning pattern of cytokine-driven inflammation.
- **The expected infection markers contributed almost nothing.** Blood cultures, urine cultures, pneumonia on imaging, and new bacterial, viral or fungal infection, the findings clinicians would rely on, were positive in 1% to 5% of patients before the diagnosis date, because cultures had not yet resulted. The model compensated with pre-infusion history and baseline labs, which is precisely the gap a decision-support tool has to fill.

LIMITATIONS

- Single-center data; no external validation at other CAR-T centers.
- Labels derive from diagnosis codes, which may lag or misrepresent the clinical picture.
- Only 80 sepsis events, limiting model complexity and precision.
- Vital signs and antimicrobial administration were not included as features.
- Features are aggregated across the pre-diagnosis window rather than resolved hour by hour.

CONCLUSIONS AND FUTURE DIRECTIONS

- Routinely collected pre-diagnosis EHR data carries meaningful signal for separating sepsis from CRS after CAR-T infusion.
- Threshold selection lets a deployed model favor sensitivity for triage or specificity for antimicrobial stewardship, depending on clinical intent.
- Next steps: external validation, addition of vital signs and time-resolved laboratory trajectories, and prospective silent-mode evaluation before any clinical deployment.